

Are you a whiz at processing information and visualising graphical representation of data? A career in GIS could be right up your street

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Think maps, and we'll bet what first pops up in your mind are all those humongous, dusty, tattered maps from your eighth standard geography class. Most of these paper maps carry all the information right there on the surface. But now picture a digital map containing all necessary information in a database—for instance, a city map packed with information on its transport network, census information or even its sewage plan. The digital map integrates all this data in different layers within the same map. So you can do away with hunting for the right map to find what you want. Simply select the information you want

to view and a map representing the requisite data will be generated accordingly. Behind such dynamic and versatile digital maps are programmers, data architects and analysts who use GIS or Geographic Information Systems. GIS is a set of powerful tools and processes to capture, manipulate, analyse and display spatial or geographically referenced data. Today, GIS is being projected as a field that holds enormous potential for growth in the days to come in the Indian IT sector.

So what's GIS all about?

GIS isn't simply 'mapping software' nor is it the same as cartography. Cartography has just one aspect to it—using data to design a map. There are two distinct aspects to GIS: one has to do with the graphical output (the map or screen display) and the other with the database, which contains all the data. A database used in GIS contains both spatial (geographic) and attribute (statistical) data. With this database attached to a map, you can generate a map carrying the information you want based on a 'geo-query'. A geo-query works like a search engine, usually involving clicking your mouse anywhere on a map with all data for that field instantly popping up.

GIS is flexible enough to map any kind of terrain, not just physical geography, right from mapping the human body to a graphical representation of instructions to assemble machine parts. And all this on the basis of the available data.

GIS is a field encompassing various other disciplines such as cartography, CAD, remote sensing, photogrammetry (aerial photos), operations research, surveying techniques and other allied spatial data processing technologies.

There's no simple or clear-cut methodology involved in GIS work processes—it differs from

Mapping your career the GIS way

